

CSA0315 – DATA STRUCTURES

CO3 – AT3

TREE TRAVERSAL CHALLENGES

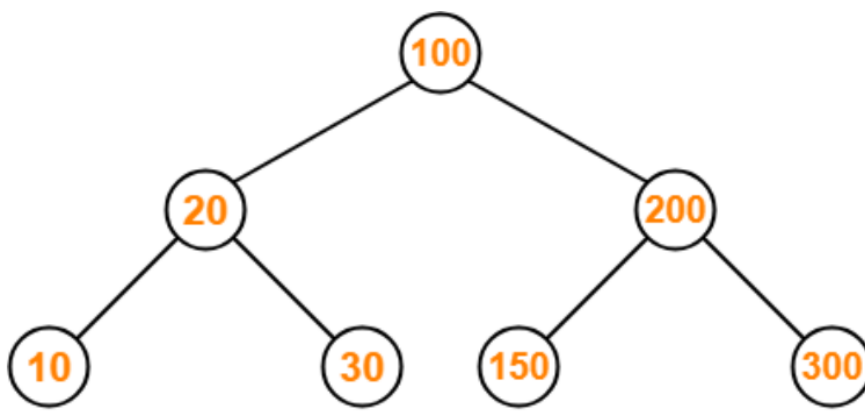
By:

HARSHITH G

CO3- AT3- Tree Traversal Challenges

Write the traversal sequences for following trees

Question No :1



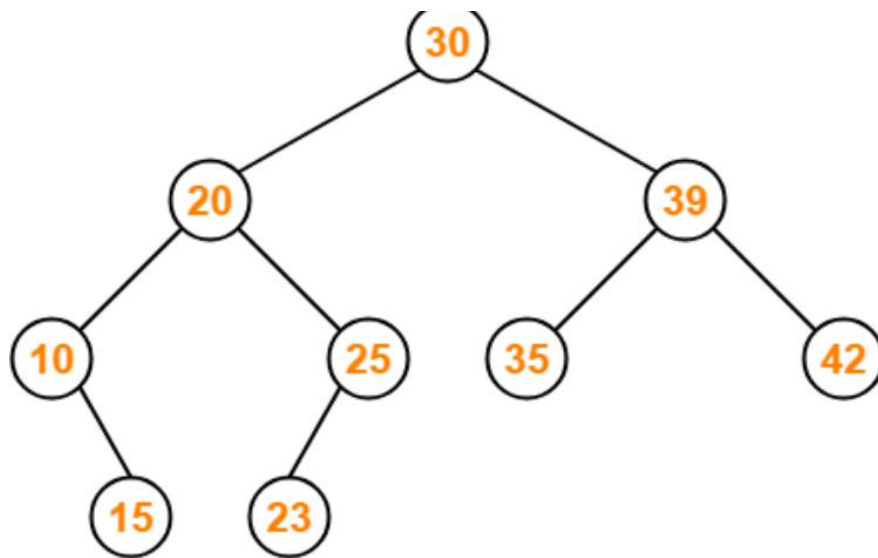
ANS:

INORDER: 10,20,30,100,150,200,300

PREORDER: 100,20,10,30,200,150,300

POSTORDER: 10,30,20,150,300,200,100

Question No : 2



ANS:

INORDER: 10,15,20,23,25,30,35,39,42

PREORDER: 30,20,10,15,25,23,39,35,42

POSTORDER: 15,10,23,25,20,35,42,39,30

Question No : 3

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

ANS:

INORDER: 0,1,2,3,4,5,6,7,8,9

PREORDER: 7,5,1,0,3,2,4,6,8,9

POSTORDER: 0,2,4,3,1,6,5,9,8,7

Question No : 5



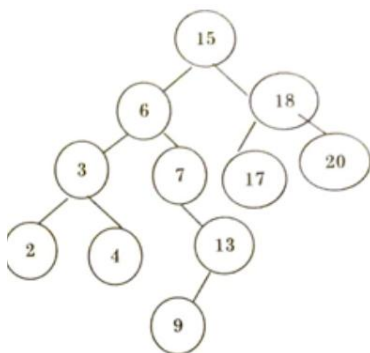
ANS:

INORDER: 4,2,7,5,1,3,6

PREORDER: 1,2,4,5,7,3,6

POSTORDER: 4,7,5,2,6,3,1

Question No 6



ANS:

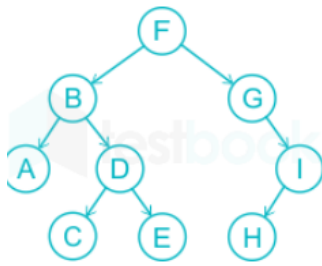
INORDER: 2,3,4,6,7,9,13,15,17,18,20

PREORDER: 15,6,3,2,4,7,13,9,18,17,20

POSTORDER: 2,4,3,9,13,7,6,17,20,18,15

Question No 7

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G

2. A, C, D, E, B, F, G, H, I

3. A, B, C, D, E, F, G, H, I

4. I, H, G, F, E, D, C, B, A

ANS: OPTION 3: A,B,C,D,E,F,G,H,I

Question No 8

Consider the following Inorder and Preorder traversal of a tree

Inorder – D B E F A G H C

Preorder - A B D E F C G H

What is the Postorder Traversal of the same tree

1. D F E B H C G A

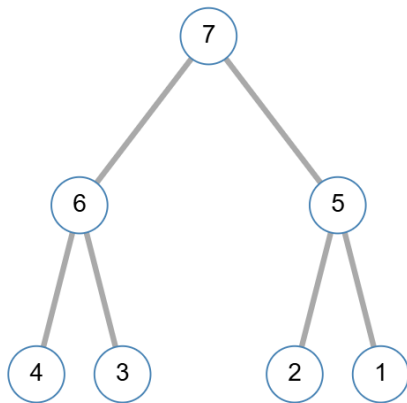
2. D F E B G H C A

3. D F E B H G C A

4. D F E H B G C A

ANS : OPTION 2. D,F,E,B,G,H,C,A

Question No 9: Find Inorder, Preorder and Postorder



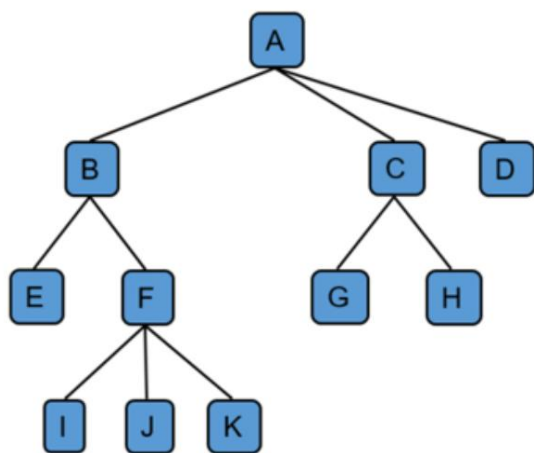
ANS:

INORDER: 4,6,3,7,2,5,1

PREORDER: 7,6,4,3,5,2,1

POSTORDER: 4,3,6,2,1,5,7

Question No 10



ANS:

PREORDER: A,B,E,F,I,J,K,C,G,H,D

POSTORDER: E,I,J,K,F,B,G,H,C,D,A

LEVEL ORDER: A,B,C,D,E,F,G,H,I,J,K